Danielle Tsao

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EDUCATION

University of Washington — **Seattle, WA, USA**
PhD, Statistics — Sept 2023 – present

University of Victoria — **Victoria, BC, Canada**
B.Sc. Honours, Statistics (GPA 8.98/9.0) — Sept 2019 – May 2023

PREPRINTS AND PUBLICATIONS

- **Danielle Tsao,** Krikamol Muandet, Frederick Eberhardt, Emilija Perković (2026). "Lost in Aggregation: The Causal Interpretation of the IV Estimand." *arXiv:2601.12120. Submitted.*
- **Danielle Tsao**, Ronan Perry, and Carlos Cinelli (2025). "On the minimum strength of (unobserved) covariates to overturn an insignificant result." *arXiv:2408.13901. To appear in Statistical Science.*
- Xiangling Ji, **Danielle Tsao**, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang (2023). "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data." *Bioinformatics Advances*, *3*(1), p.vbad030. DOI: 10.1093/bioadv/vbad030

RESEARCH EXPERIENCE

Research Assistant, Statistics, University of Washington — 2025-2026
Supervisor: Emilija Perković (University of Washington)
- Studied the causal interpretation of the instrumental variable estimand under an aggregate treatment.

Research Assistant, Statistics, University of Washington — 2024-2025
Supervisor: Carlos Cinelli (University of Washington)
- Studied reversals of insignificance and robust statistical inference of linear regression coefficients.

Research Assistant, Biostatistics, Mitacs Accelerate Internship Program — 2022-2023
Supervisor: Xuekui Zhang (University of Victoria)
- Led a research team through the development of a deconvolution benchmarking pipeline in R.

Research Assistant, Biostatistics, University of Victoria — 2022
Supervisor: Xuekui Zhang
- Led a summer research team through the development of a deconvolution benchmarking pipeline in R.

TEACHING

Teaching Assistant, Statistics, University of Washington — 2023-2026
- STAT 564: Bayesian Statistics for the Social Sciences (Spring 2026)
- STAT 502: Design and Analysis of Experiments (Winter 2025)
- STAT 311: Elements of Statistical Methods (Summer 2025)
- STAT 395: Probability II (Winter 2024)
- STAT 220: Statistical Reasoning (Fall 2023, Summer 2024)

Teaching Assistant, Mathematics and Statistics, University of Victoria — 2022
- Tutored and graded undergraduate math and statistics courses in the Math and Stats Assistance Centre.

ACADEMIC SERVICE

- **Workflow Chair** for the 42nd Conference on Uncertainty in Artificial Intelligence 2026
- **Mentor** for the Statistics Pre-application Review Service, University of Washington 2025-present
- **Volunteer Consultant** for the Tribal-UW program, Tribal Exchange Network 2025-present
- **Member** of the Statistics Outreach and Community Committee, University of Washington 2024-present
- **Peer mentor** for Statistics PhD program, University of Washington 2024-present

PRESENTATIONS AND RESEARCH STAYS

- **Causal inference: From theory to practice and back again (CIF),** Cambridge, UK. Hosted by the Isaac Newton Institute of Mathematical Sciences. Invited Participant.
- **CIF Workshop on Foundations of Causal Inference**, Cambridge, UK, "Lost in Aggregation: The Causal Interpretation of the IV Estimand", joint work with Emilija Perković (advisor), Krikamol Muandet and Frederick Eberhardt, Poster Presentation.
- **Jamie Cassels Undergraduate Research Award Poster Fair 2023,** Victoria, Canada. "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data", joint work with Xiangling Ji, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang, Poster Presentation.
- **Canadian Statistics Student Conference (CSSC) 2022**, "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data", joint work with Xiangling Ji, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang, Poster Presentation.

HONOURS AND AWARDS

Natural Sciences and Engineering Research Council of Canada 2024-present
Canada Graduate Research Scholarship, Doctoral program ($120,000)

American Statistical Association
Gertrude M. Cox Scholarship Honourable Mention 2025

University of Victoria
- University of Victoria Faculty of Science Jubilee Medal 2023
- University of Victoria Excellence Scholarship ($26,000) 2019-2023
- Faculty of Science Dean's List 2019-2023
- Science Emerging Researcher Award ($5,500) 2022
- Jamie Cassels Undergraduate Research Award in Mathematics and Statistics ($1,500) 2022
- Mathematics and Statistics Scholarship ($500) 2022
- NSERC Undergraduate Student Research Award in Statistics ($6,000) 2021
- Mathematics and Statistics Angus & Annie MacKay Scholarship ($1,700) 2021
- Mathematics and Statistics Donald & Evelyn Munro Scholarship ($1,000) 2019
- Excellence in Math Scholarship ($2,500) 2019

Canadian Statistics Student Conference
- Best Undergraduate Poster Presentation 2022

Canadian Mathematical Society
- CMS-RBC Excellence in Mathematics Scholarship ($1,000) 2021

Performing Arts, featured soloist with:
- Victoria Symphony – Symphony Splash 2018
- Sooke Philharmonic Orchestra – Young Masters Concert Series, first prize 2018
- Victoria Chamber Orchestra – Louis Sherman Concerto Competition, first prize 2017